

UX-0006 — Butlerly AI Interaction and Explainability Specification

Version 1.0

Purpose

This document defines how Butlerly uses AI in Version 1, how AI behavior is presented to users, when AI may act automatically, when confirmation is required, how explanations are structured, how privacy and provider boundaries are disclosed, and how users correct, undo, or disable AI behavior. It governs all AI-assisted capture, extraction, classification, matching, search, explanation, recommendation, and supported action flows.

1. AI Experience Objective

Butlerly should make routine personal-finance organization feel effortless without making the user surrender understanding or control.

The AI experience must:

- Reduce repetitive data entry and review.
- Prefer useful proposals over blank forms.
- Handle high-confidence, low-consequence work automatically.
- Surface only meaningful exceptions.
- Explain material decisions in plain language.
- Distinguish fact, inference, and recommendation.
- Preserve source evidence and prior values.
- Allow correction, undo, and recovery.
- Respect local-first processing and explicit consent for external processing.
- Remain provider-independent.
- Degrade gracefully when AI is unavailable.

2. Governing Principles

2.1 AI Is a Capability, Not the Product Structure

The Assistant is an optional, non-primary capability in Finance Version 1. AI may also appear contextually where enabled and approved, but core navigation and core workflows must remain complete without AI. Users should not need to open a chatbot to benefit from optional extraction, classification, duplicate detection, search assistance, or explanations.

2.2 Propose Before Asking

When Butlerly has enough evidence to suggest a value, it should present the suggestion rather than asking the user to start from an empty field.

2.3 Confidence and Consequence Determine Autonomy

AI autonomy is based on both confidence and the potential consequence of being wrong. High confidence alone is insufficient for an irreversible or high-impact action.

2.4 Explain Material Decisions

Routine background work may remain quiet. A decision becomes materially explainable when it changes a record, affects money-related interpretation, creates or removes a relationship, sends data externally, or asks the user to act.

2.5 User Correction Is Authoritative

A user correction overrides the AI result. Butlerly must not repeatedly reapply a rejected interpretation unless new evidence materially changes the situation.

2.6 Privacy Is Part of the Interaction

Before external processing, Butlerly must explain what data leaves the device, which provider receives it, why it is needed, what alternatives exist, and whether the result is retained.

2.7 No False Certainty

Butlerly must state uncertainty honestly. It must not convert weak evidence into confident language or present a recommendation as a fact.

3. AI Capability Classes

A01 — Extraction

Reads receipt images, files, statements, and structured imports to identify merchant, date, amount, tax, payment source, line items, and other supported fields.

A02 — Normalization

Standardizes merchant names, dates, currencies, categories, and source labels while preserving original values.

A03 — Classification

Suggests categories, labels, record types, and import mappings.

A04 — Matching and Linking

Suggests receipt-to-transaction links, source-document relationships, merchant identity, and duplicate candidates.

A05 — Search Interpretation

Converts natural-language queries into local search filters and retrieves relevant records.

A06 — Explanation

Explains why Butlerly categorized, matched, linked, flagged, or recommended something.

A07 — Recommendation

Suggests a next action such as confirm, correct, merge, defer, reconnect, or inspect source.

A08 — Supported Action Initiation

Prepares an action requested through the Assistant, but follows confirmation requirements before execution.

A09 — Learning from Corrections

Uses approved correction patterns to improve future proposals without hiding what was learned or removing reset controls.

4. Autonomy Model

Level 0 — No AI Action

Butlerly displays source data and waits for direct user input.

Use when evidence is insufficient, AI is disabled, or the task is outside approved capability.

Level 1 — Suggest

Butlerly proposes a value or action but does not apply it.

Use for medium-confidence interpretations, new behavior patterns, or meaningful choices.

Level 2 — Apply with Immediate Undo

Butlerly applies a reversible, low-consequence change and provides Undo and version history.

Examples: merchant normalization, a high-confidence category, or a clear receipt-to-transaction link.

Level 3 — Apply Quietly with Traceability

Butlerly completes high-confidence, routine, reversible work without interrupting the user. The reason, provider, timestamp, and prior value remain available in detail or history.

Level 4 — Explicit Confirmation Required

Butlerly prepares the action and requires confirmation.

Use for destructive, externally visible, security-sensitive, privacy-sensitive, irreversible, or materially consequential actions.

AI must never independently:

- Permanently delete data.
- Erase all user data.
- Revoke a trusted device.
- Enable external data processing without consent.
- Send or share financial data to another person.
- Commit a transaction or payment.
- Change security credentials.
- Override an explicit user correction silently.

5. Confidence Model

User-facing confidence uses three labels:

High

Evidence strongly supports the interpretation and common alternatives are unlikely.

Medium

The interpretation is plausible but another reasonable value may exist.

Low

Evidence is incomplete, inconsistent, ambiguous, or dependent on unsupported assumptions.

Confidence must be calibrated per capability. A 90% extraction confidence is not automatically equivalent to a 90% duplicate confidence.

Internal numerical scores may be used for ranking and testing, but the primary UI should use understandable labels unless a detailed diagnostic view is explicitly needed.

Confidence presentation rules:

- Show confidence when the user is asked to review or rely on an inference.
- Do not display confidence as decorative precision.
- Never use color alone.
- Pair Low or Medium confidence with the missing or conflicting evidence.
- Do not imply that High means guaranteed.

6. Consequence Model

Low consequence

Easy to reverse; limited effect on organization or retrieval.

Examples: label suggestion, merchant display normalization.

Medium consequence

Could affect reports, search, relationships, or future automation.

Examples: category, receipt-to-transaction link, duplicate decision.

High consequence

Could cause data loss, privacy exposure, security impact, or a materially misleading financial conclusion.

Examples: permanent deletion, device revocation, external sharing, broad merge, erase all data.

Decision rule:

- High confidence + low consequence → may apply automatically.
- High confidence + medium consequence → apply with Undo or request confirmation depending on context.
- Medium confidence + low consequence → suggest or queue for review.

- Medium confidence + medium consequence → require review.
- Any confidence + high consequence → explicit confirmation and authentication where required.
- Low confidence → ask one focused question, preserve source, or defer.

7. Standard Explanation Structure

Every material AI explanation must answer:

1. What Butlerly believes

A direct, concise conclusion.

2. Why Butlerly believes it

The strongest relevant evidence, not a hidden chain of thought.

3. Confidence or uncertainty

High, Medium, or Low, plus the reason for uncertainty when useful.

4. What Butlerly did or recommends

State whether the value was applied, proposed, deferred, or blocked.

5. What the user can do

Confirm, correct, inspect source, undo, defer, or ask a follow-up.

Example:

“Butlerly categorized this as Dining because the receipt is from a restaurant, the merchant matched prior Dining records, and the line items appear to be meals. Confidence: High. The category was applied automatically. You can change it or undo.”

Explanation content must not expose private implementation prompts, hidden reasoning traces, security-sensitive rules, model secrets, or unsupported speculation.

8. Fact, Inference, and Recommendation

Observation

Directly supported by source data.

Example: “The receipt total is \$42.18.”

Inference

Derived from evidence and may be uncertain.

Example: “This appears to be a restaurant purchase.”

Recommendation

A proposed next action.

Example: “Link this receipt to the matching card transaction.”

The interface must not blur these categories. When an answer contains more than one, wording and visual hierarchy should make the distinction clear.

9. AI Interaction Patterns

9.1 Inline Proposal

Used inside a field or detail view.

Required elements: proposed value, confidence when material, brief reason, accept/edit control.

9.2 Automatic Completion with Undo

Used after capture or import.

Required elements when surfaced: what was completed, Undo, open record, review status.

9.3 Review Card

Used in Review.

Required elements: issue, recommendation, evidence, confidence, primary action, alternatives, defer.

9.4 Contextual Explanation

Opened from a field, relationship, match, or record.

Required elements: conclusion, evidence, uncertainty, source links, correction.

9.5 Conversational Answer

Used in Global or Contextual Assistant.

Required elements: direct answer first, supporting evidence, uncertainty, relevant record links, next action.

9.6 Focused Clarification

Ask one concrete question at a time. Offer likely answers when available. Allow defer when the missing information is not blocking.

9.7 Action Preview

Before an AI-initiated action that requires confirmation, show exactly what will change, what remains unchanged, and whether Undo is available.

10. Capture and Extraction Behavior

For receipt or document intake, Butlerly must:

- Preserve the original source before interpretation.
- Identify processing location: local or external.
- Extract supported fields and retain field-level confidence.
- Avoid blocking the user for nonessential fields.

- Ask at most one focused blocking question at a time.
- Save incomplete records when safe and route them to Review.
- Expose the source near uncertain extracted values.
- Never discard the original because extraction failed.

Field behavior:

- High-confidence field: populate automatically.
- Medium-confidence field: populate as proposal or mark for review.
- Low-confidence required field: ask one question.
- Low-confidence optional field: leave unset or defer.

11. Classification and Matching Behavior

Category proposals should consider approved context such as merchant, source text, prior user corrections, record type, and line items.

Matching proposals should show:

- Records being compared.
- Matching evidence such as amount, date, merchant, and source.
- Meaningful differences.
- Recommended relationship.
- Confidence.
- Merge, link, keep separate, or inspect actions as applicable.

Duplicate resolution must never silently delete a source artifact. Merging creates a canonical record while preserving original evidence and history.

12. Assistant Behavior

The Assistant supports:

- Questions about records.
- Search assistance.
- Explanations of Butlerly decisions.
- Supported action preparation.
- Product guidance relevant to the current task.

The Assistant must:

- Preserve current context when invoked from a record.
- Use only approved related data.
- State when the answer is based only on local data.
- State when unavailable sources may affect completeness.
- Provide record links or evidence rather than unsupported assertions.
- Ask clarification only when materially necessary.
- Avoid long conversational detours when a direct product action is available.

- Never claim an action completed unless the system confirms completion.

The Assistant must not:

- Replace stable navigation.
- Invent records or values.
- Present financial, legal, or tax conclusions beyond supported product scope.
- Hide provider use.
- Pressure the user to enable external AI.
- use anthropomorphic language to imply consciousness, emotion, or independent authority.

13. Supported Action Confirmation

An Assistant request may prepare an action such as:

- Change a category.
- Link records.
- Start an import.
- Create an export.
- Open device settings.
- Defer a review item.

Action preview must include:

- Action name.
- Affected records or settings.
- Proposed changes.
- Reversibility.
- External processing or sharing, if any.
- Confirm and Cancel.

The UI must wait for verified system completion before stating success.

14. External AI Consent

Consent must be contextual and feature-specific where practical.

The consent interface must state:

- Exact or understandable data scope.
- Provider identity.
- Purpose.
- Whether identifiers are removed or minimized.
- Retention behavior known to Butlerly.
- Whether the result is stored locally.
- Local or manual alternative.
- Available choices: Allow Once, Allow for This Feature, Not Now.

Consent rules:

- No preselected approval.
- Denial must not disable unrelated local functionality.
- Users can review and revoke consent in Privacy & Data.
- Provider changes require renewed disclosure when the boundary materially changes.
- Sensitive previews must avoid showing unnecessary private data.

15. Local AI and Provider Independence

The user experience should describe capability and processing location rather than expose architecture unnecessarily.

Preferred language:

- “Processed on this device.”
- “External processing is required for this step.”
- “This feature is temporarily unavailable; your receipt is saved.”

Avoid primary UI language such as model endpoint, inference server, token window, vector database, or provider adapter.

A provider switch must not change the fundamental interaction model, explanation structure, consent standard, or record history.

16. Learning from User Corrections

Learning is optional and user-controlled.

Butlerly may learn patterns such as:

- Merchant-to-category preference.
- Preferred normalized merchant name.
- Repeated receipt-to-account relationship.
- Import mapping preference.

Butlerly must not learn:

- A rule that contradicts explicit policy or safety limits.
- Sensitive inference beyond approved scope.
- Hidden cross-user patterns presented as personal preferences.

When learning is enabled:

- Show that a correction may improve future suggestions.
- Allow the user to apply only once.
- Provide Learned Preferences in Settings.
- Allow individual removal and full reset.
- Preserve original history even if a learned preference changes future results.

17. Correction, Undo, and Recovery

Correction paths must be available from:

- Inline proposal.
- Review item.
- Record Detail.
- Search result where appropriate.
- Assistant answer.
- Version History.

On correction, Butlerly must:

- Save the user-selected value as authoritative.
- Record prior value, source, device, and timestamp.
- Update dependent views.
- Preserve the correction locally. Future optional synchronization may propagate it when that capability exists and is enabled.
- Ask whether to learn only when useful and not repetitive.

Undo is the immediate convenience path. Version History is the durable recovery path.

18. Failure and Degraded Behavior

When AI processing fails, the interface must state:

- What failed.
- What was preserved.
- Whether data left the device.
- What remains usable.
- What the user can do next.
- Whether retry is safe.

Recovery options may include:

- Retry local processing.
- Retry external processing with consent.
- Use another approved provider.
- Complete manually.
- Save and review later.
- Inspect the original source.

AI failure must not block access to locally stored records or unrelated features.

19. AI States and Required Screens

S-AI-001 Global Assistant

Default, empty, local-only, offline, provider unavailable, context unavailable.

S-AI-002 Contextual Record Assistant

Selected-record context, related-context available, source missing, deleted/read-only record.

S-AI-003 AI Answer

Complete answer, partial answer, uncertainty, no evidence, unsupported request, action proposed.

S-AI-004 AI Processing Consent

First use, allow once, persistent feature consent, provider changed, consent revoked.

S-AI-005 AI Unavailable

Local model unavailable, external provider unavailable, network unavailable, rate/usage limit, unsupported format.

S-AI-006 Learned Preferences

Enabled, disabled, empty, populated, individual preference detail, reset confirmation.

S-INB-002 Review Item

AI proposal, confidence, evidence, confirm, correct, defer.

S-CAP-005 Focused Clarification

Single question, suggested answers, manual response, skip/review later.

20. Device Adaptation

Smartphone

- Short, direct explanations.
- One review decision at a time.
- Bottom-sheet consent and action previews when appropriate.
- Voice input where supported.
- Source preview accessible without losing the task.

Tablet

- List-detail review.
- Side-by-side source and interpreted record.
- Assistant in secondary pane where space allows.
- Keyboard, pointer, and touch support.

Desktop

- Multi-pane evidence, record, and explanation.
- Keyboard-first review and correction.
- Safe batch review only for genuinely equivalent proposals.

- More visible diagnostics and provider settings without exposing unnecessary complexity.

21. Content and Tone Standard

AI language should be:

- Calm.
- Direct.
- Specific.
- Humble about uncertainty.
- Free of hype.
- Free of blame.
- Focused on the user's next decision.

Preferred:

"Butlerly could not read the date. The receipt image is saved. Enter the date or review this later."

Avoid:

"AI failed unexpectedly."

"Something magical happened."

"I know this is Dining."

"You must enable cloud AI to continue."

Use "Butlerly" for product behavior and "Assistant" for the conversational surface. Do not overuse "AI" in routine UI copy.

22. Accessibility Requirements

- Every confidence label has text, not color alone.
- Evidence lists have logical reading order.
- Source images include accessible descriptions and extracted-text alternatives where possible.
- Dynamic processing and completion states are announced appropriately.
- Streaming answers do not create unusable repeated announcements.
- Keyboard users can open evidence, correct, confirm, defer, and undo.
- Large text reflows explanation cards without truncation.
- Reduced motion disables decorative generation or progress effects.
- Focus returns to the initiating control after consent or action dialogs.
- Suggested prompts are accessible controls, not placeholder-only text.

23. Privacy-Preserving Analytics

Permitted flow-level events may include:

- AI capability invoked.
- Local or external processing selected.
- Consent allowed once, allowed for feature, or declined.

- Proposal accepted, corrected, deferred, or ignored.
- Undo used.
- Explanation opened.
- Source inspected.
- Provider unavailable.
- Manual fallback selected.

Analytics must not capture receipt contents, financial values, user questions, model prompts, model responses, or private record data by default.

24. Quality and Safety Evaluation

Each AI capability must be evaluated for:

- Accuracy.
- Confidence calibration.
- False-positive and false-negative impact.
- Explanation faithfulness to available evidence.
- Privacy boundary compliance.
- Reversibility.
- Accessibility.
- Provider failure behavior.
- Cross-device consistency.
- Performance on representative formats and languages.

Testing must include:

- Ambiguous receipts.
- Poor images.
- Conflicting source values.
- Duplicate candidates.
- Offline mode.
- External consent denied.
- Provider unavailable.
- User correction followed by a similar record.
- Same record viewed across device form factors; cross-device synchronization testing applies only when that future capability is implemented.
- Unsupported questions and actions.

25. Implementation Metadata

For material AI-generated or AI-modified values, the system should retain appropriate metadata:

- Capability type.
- Model or provider identifier at an implementation-appropriate level.
- Processing location.
- Timestamp.

- Input source references.
- Confidence.
- Applied, suggested, confirmed, corrected, or rejected status.
- Prior value.
- User correction.
- Consent basis when external processing was used.

This metadata supports explanation, version history, debugging, audits, and provider migration. It should not clutter the default UI.

26. Dependencies and Governance

UX-0006 depends on:

- UX-0001 — Butlerly Experience Blueprint.
- UX-0002 — Butlerly Version 1 User Journey Map.
- UX-0003 — Butlerly Information Architecture and Navigation Model.
- UX-0004 — Butlerly Core User Flows.
- UX-0005 — Butlerly Screen and State Inventory.
- Product MVP scope and personas.
- Architecture decisions for AI abstraction, local processing, privacy, encryption, and audit metadata.

UX-0006 governs:

- UX-0007 — Design System Foundations.
- UX-0008 — Cross-Device Experience and Continuity.
- UX-0009 — Accessibility Standard.
- UX-0010 — Low-Fidelity Wireframe Specification.
- AI-related engineering stories, prompt/output contracts, provider adapters, analytics, QA, and usability testing.

27. Acceptance Criteria

This specification is complete when:

- Every Version 1 AI capability has a defined interaction pattern.
- Autonomy is determined by confidence, consequence, reversibility, and consent.
- Material AI decisions use the standard explanation structure.
- Observation, inference, and recommendation are distinguishable.
- Users can inspect evidence, correct, undo, defer, and recover.
- External processing is never hidden or enabled without appropriate consent.
- Local-only and provider-failure paths remain usable.
- Smartphone, tablet, and desktop behavior is defined.
- Accessibility requirements are testable.
- Engineering can retain the metadata required for explanation and history.
- QA can test accuracy, calibration, privacy, recovery, and provider independence.

BDS-0000 Conformance

BDS-0000 — Butler Design System v1.0 governs the visual and implementation presentation of AI states, including semantic tokens, theme values, typography, spacing/radius, responsive breakpoints, component/state behavior, accessibility, localization/original-data handling, and Flutter UI constraints. Finance V1 visible primary navigation is Home, Transactions, Review, Search, and Settings; Assistant remains optional/non-primary. AI must not introduce a parallel navigation model. Core operation requires no Butlerly account. Device trust and synchronization remain future/optional unless product scope explicitly enables them. UI-authored AI explanations follow the selected application language while quoted, scanned, imported, or otherwise source-derived real data preserves its original language/script by default; original financial amount/currency is never replaced by a converted reference value.